Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

WEEK-2

2520030078

M.Apoorva

Section – 3

Step 1 — Create Database

```
CREATE DATABASE IF NOT EXISTS bookflow4_db;
```

Ouput:

```
1  14:56:00  CREATE DATABASE IF NOT EXISTS bookflow4_db
```

Use database:

```
USE bookflow4_db;
```

Output:

```
2  14:57:30  USE bookflow4_db
```

Step 2 — Create Books Table (DDL + Constraints)

```
CREATE TABLE Books (
 book_id INT PRIMARY KEY,
 title VARCHAR(100) NOT NULL,
 isbn VARCHAR(20) UNIQUE,
 published_year INT CHECK (published_year < 2027)
);
```

Output:

```
3  14:58:15  CREATE TABLE Books ( book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VAR
```

Select:

```
SELECT * FROM Books;
```

Output:

book_id	title	isbn	published_year
NULL	NULL	NULL	NULL

Step 3 & 4 — Insert Books and Display

```
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, '1984', '9780451524935', 1949);
SELECT * FROM Books;
```

Output:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Step 5, 6 & 7 — Members Table: Create, Insert, Display

```
CREATE TABLE Members (
 member_id INT PRIMARY KEY,
 full_name VARCHAR(100),
 email VARCHAR(100) UNIQUE
);
SELECT * FROM Members;
```

Output:

member_id	full_name	email
NULL	NULL	NULL

Insert:

```
INSERT INTO Members (member_id, full_name, email) VALUES
(101, 'Sushma', 'sushma@email.com'),
(102, 'Sowmya', 'sowmya@email.com'),
(103, 'Mokshagna', 'mokshagna@email.com');
SELECT * FROM Members;
```

Output:

member_id	full_name	email
101	Sushma	sushma@email.com
102	Sowmya	sowmya@email.com
103	Mokshagna	mokshagna@email.com
NULL	NULL	NULL

Step 8 — Create Loans Table (FOREIGN KEYS)

```
CREATE TABLE Loans (
 loan_id INT PRIMARY KEY,
 member_id INT,
 book_id INT,
 loan_date DATE,
 FOREIGN KEY (member_id) REFERENCES Members(member_id),
 FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
SELECT * FROM Loans;
```

Output:

loan_id	member_id	book_id	loan_date
NULL	NULL	NULL	NULL

Step 9 & 10 — Insert 10 Loans and Display

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
SELECT * FROM Loans;
```

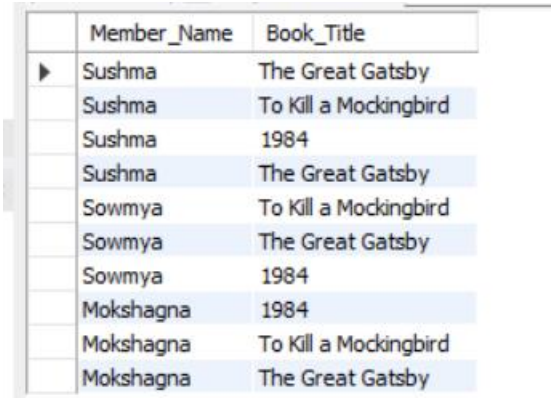
Output:

loan_id	member_id	book_id	loan_date
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01
NULL	NULL	NULL	NULL

Step 11 — JOIN Query (Who Borrowed What)

```
SELECT m.full_name AS Member_Name, b.title AS Book_Title
FROM Loans l
INNER JOIN Members m ON l.member_id = m.member_id
INNER JOIN Books b ON l.book_id = b.book_id;
```

Output:

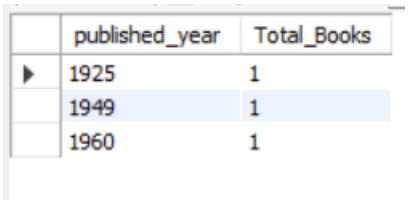


Member_Name	Book_Title
Sushma	The Great Gatsby
Sushma	To Kill a Mockingbird
Sushma	1984
Sushma	The Great Gatsby
Sowmya	To Kill a Mockingbird
Sowmya	The Great Gatsby
Sowmya	1984
Mokshagna	1984
Mokshagna	To Kill a Mockingbird
Mokshagna	The Great Gatsby

Step 12 — GROUP BY Query (Books per Year)

```
SELECT published_year, COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year
ORDER BY published_year;
```

Output:



published_year	Total_Books
1925	1
1949	1
1960	1

Step 13–16 — Donation via TRANSACTION (Atomicity)

```
CREATE TABLE Donation_History (
 donation_id INT PRIMARY KEY,
 book_id INT,
 donor_name VARCHAR(100),
 donation_date DATE,
 FOREIGN KEY (book_id) REFERENCES Books(book_id)
);
START TRANSACTION;
INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);
INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());
COMMIT;


SELECT * FROM Books;
```

Output:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	Animal Farm	9780451526342	1945
NULL	NULL	NULL	NULL

```
SELECT * FROM Donation_History;
```

Output:

donation_id	book_id	donor_name	donation_date
1	4	Raj Kumar	2026-08-10
NULL	NULL	NULL	NULL

Step 17 & 18 — Index on ISBN + Fast Search

```
CREATE INDEX idx_books_isbn ON Books(isbn);
SELECT * FROM Books WHERE isbn = '9780451524935';
```

Output:

book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

----------X----------